

WIGHT ST CATHERINES POINT, United Kingdom

WMO: 038660

Lat: 50.5756N

Lon: 1.2969W

Elev: 24

StdP: 101.04

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.8	0.5	-5.5	2.4	1.4	-4.0	2.7	2.1	18.9	8.9	16.7	9.0	5.0	50	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.2	22.9	18.1	21.4	17.3	20.2	16.7	18.7	21.7	18.0	20.3	17.4	19.3	5.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.7	19.8	17.1	12.2	19.1	16.6	11.8	18.6	53.2	22.0	50.9	20.5	49.1	19.5	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	14.1	12.6	-1.8	25.4	1.9	1.7	-3.2	26.6	-4.3	27.6	-5.4	28.5	-6.8	29.8		
(5)			-2.9	20.1	1.8	0.8	-4.2	20.6	-5.3	21.1	-6.3	21.6	-7.6	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	6.7	6.4	7.5	9.4	12.1	14.8	16.9	17.4	16.0	13.5	10.1	7.7		
	DBStd	4.56	2.77	2.51	2.19	2.33	2.31	1.88	1.85	1.60	1.78	2.29	2.78	3.07		
	HDD10.0	443	104	101	79	36	5	0	0	0	0	4	32	82		
	HDD18.3	2495	361	334	336	268	193	109	52	39	74	151	247	331		
	CDD10.0	1014	1	1	2	18	71	143	216	228	178	111	36	9		
	CDD18.3	23	0	0	0	0	1	2	9	9	2	0	0	0		
	CDH23.3	31	0	0	0	0	1	3	14	12	1	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	6.0	6.7	6.0	5.8	5.1	5.4	5.4	5.6	5.8	6.0	6.5	6.4	6.9		
(15) Precipitation	PrecAvg	775	88	63	48	50	46	48	43	50	58	91	97	89		
	PrecMax	1034	189	150	137	123	125	126	113	139	133	169	194	178		
	PrecMin	591	16	6	4	0	7	5	3	4	5	26	26	26		
	PrecStd	125	43	41	26	31	29	31	27	32	34	39	49	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	11.5	14.0	18.3	21.2	23.4	25.2	24.8	22.4	19.1	15.9	13.6		
		MCWB	10.6	9.5	10.2	13.3	15.8	18.1	18.8	19.3	18.4	16.5	14.6	12.4		
	2%	DB	11.2	10.6	12.1	15.7	18.7	20.7	22.9	22.7	20.6	17.7	14.9	12.9		
		MCWB	10.2	9.3	9.4	11.9	14.4	16.6	17.8	18.5	17.3	15.7	13.6	11.8		
	5%	DB	10.7	10.0	11.0	13.9	17.0	19.1	21.3	21.2	19.4	17.0	14.3	12.3		
		MCWB	9.6	8.9	8.9	10.7	13.6	15.5	17.1	17.5	16.4	15.3	13.0	11.1		
	10%	DB	10.1	9.5	10.2	12.6	15.6	17.7	19.9	19.9	18.5	16.4	13.8	11.6		
		MCWB	9.0	8.4	8.6	10.1	12.9	14.8	16.7	16.8	16.0	14.8	12.4	10.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.3	10.9	14.0	16.8	18.6	19.6	20.0	19.0	17.2	15.2	12.8		
		MCDB	11.4	10.7	12.5	17.3	20.0	22.4	23.7	23.5	21.5	18.4	15.6	13.1		
	2%	WB	10.5	9.7	10.1	12.4	15.1	17.2	18.4	18.9	18.0	16.3	14.1	12.1		
		MCDB	10.9	10.2	11.2	15.1	17.6	20.2	21.6	22.0	19.7	17.3	14.6	12.6		
	5%	WB	9.9	9.2	9.6	11.2	14.1	16.1	17.7	18.1	17.2	15.7	13.3	11.4		
		MCDB	10.5	9.8	10.5	13.1	16.2	18.3	20.3	20.2	18.6	16.7	14.1	12.1		
	10%	WB	9.3	8.7	9.0	10.5	13.2	15.3	17.1	17.5	16.7	15.2	12.6	10.6		
		MCDB	10.0	9.3	9.9	12.0	15.1	17.0	19.2	19.2	18.0	16.2	13.6	11.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.2	3.4	3.9	4.5	4.5	4.3	4.1	4.2	4.4	3.8	3.5	3.4		
		MCDBR	2.9	3.2	4.6	6.3	6.7	6.8	6.7	6.1	5.5	3.6	3.1	2.7		
	5% WB	MCWBR	3.4	3.0	3.7	4.5	4.7	4.7	4.4	4.2	4.0	3.1	3.3	3.4		
		MCWBR	2.9	2.9	3.8	5.2	5.9	5.8	5.5	5.0	4.4	3.2	2.9	2.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.334	0.350	0.387	0.403	0.406	0.402	0.395	0.389	0.378	0.364	0.337	0.325		
		taud	2.383	2.392	2.291	2.278	2.319	2.358	2.384	2.418	2.435	2.443	2.429	2.367		
	Ebn at Noon	Ebn at Noon	687	778	808	838	852	856	857	844	813	752	680	635		
		Edh at Noon	65	82	107	121	121	118	113	104	92	77	61	57		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.83	1.57	2.81	4.50	5.42	6.00	5.69	4.78	3.52	1.97	1.02	0.65		
	RadStd	RadStd	0.06	0.18	0.35	0.40	0.32	0.54	0.42	0.35	0.25	0.16	0.11	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page